STEM basics

Intern study, week 1

2026.09.19

정수영

Scope

I will focus on the **formation of a (S)TEM image**

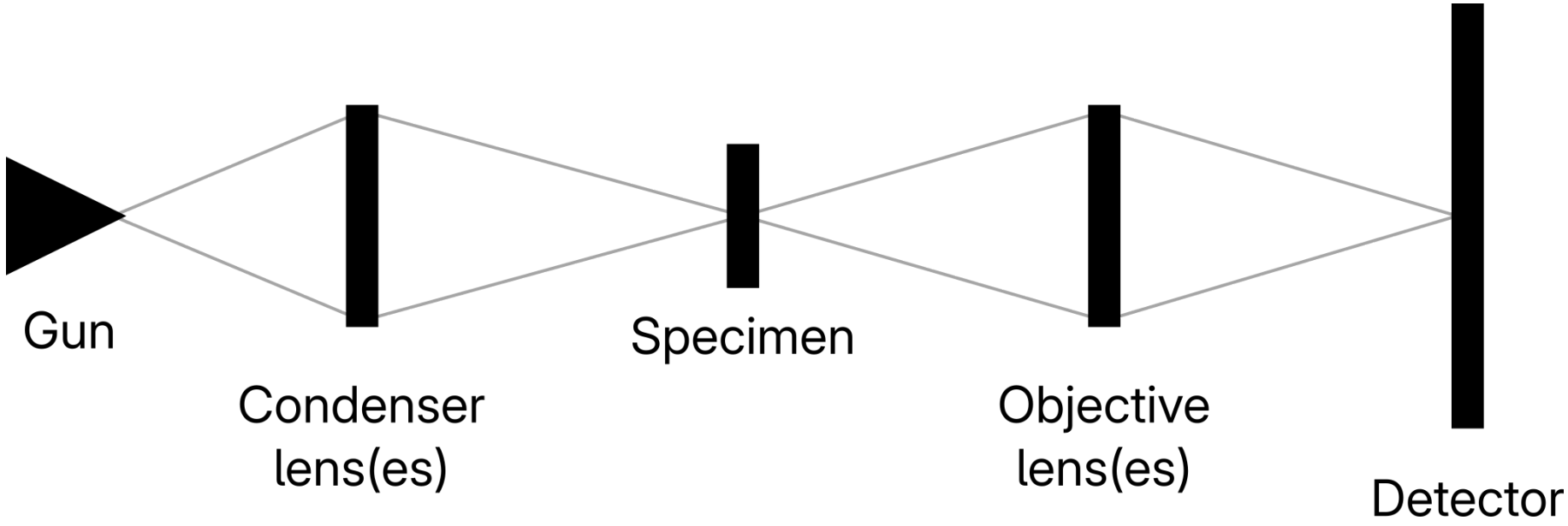
- Geometry of (S)TEM
- Electron-sample interaction
- Information on the detector

I will not cover how to *practically obtain* a high-resolution image:

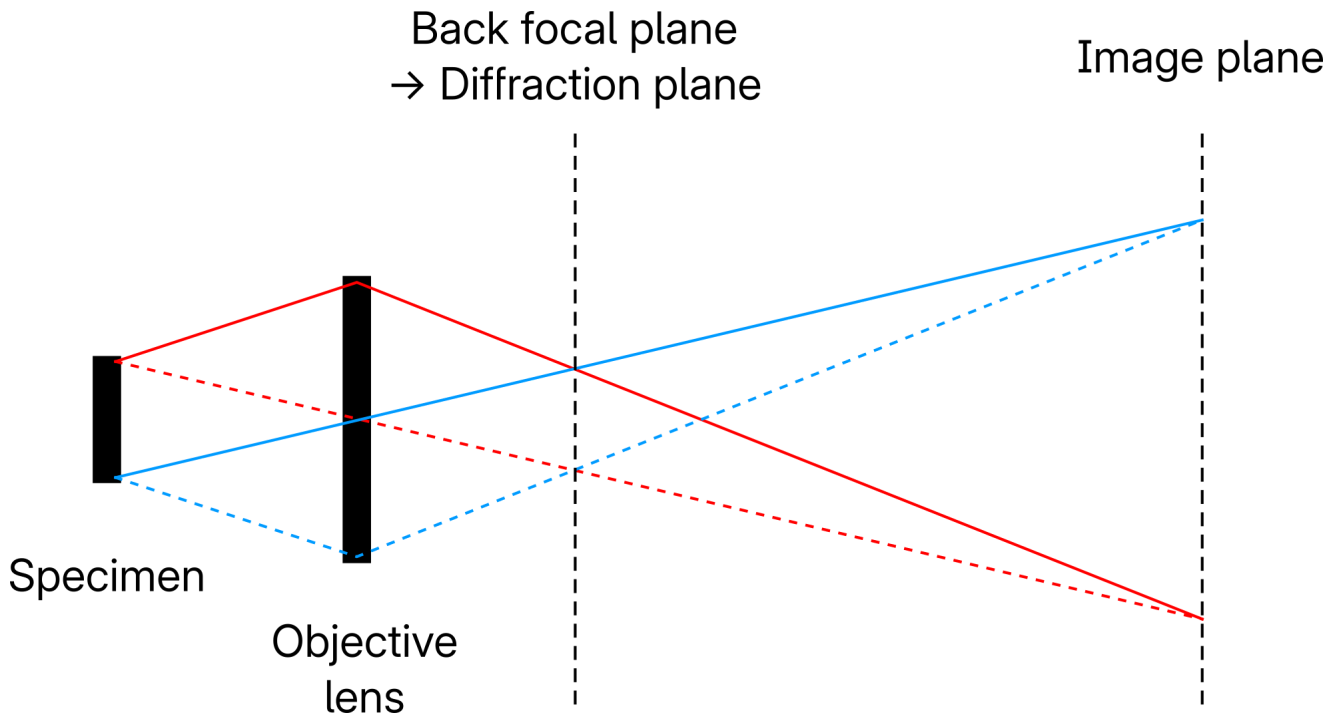
Electron guns, Aberration correction, Ronchigrams, Kikuchi lines,...

> *Lower-resolution techniques lead to higher-resolution techniques.*

Model (S)TEM

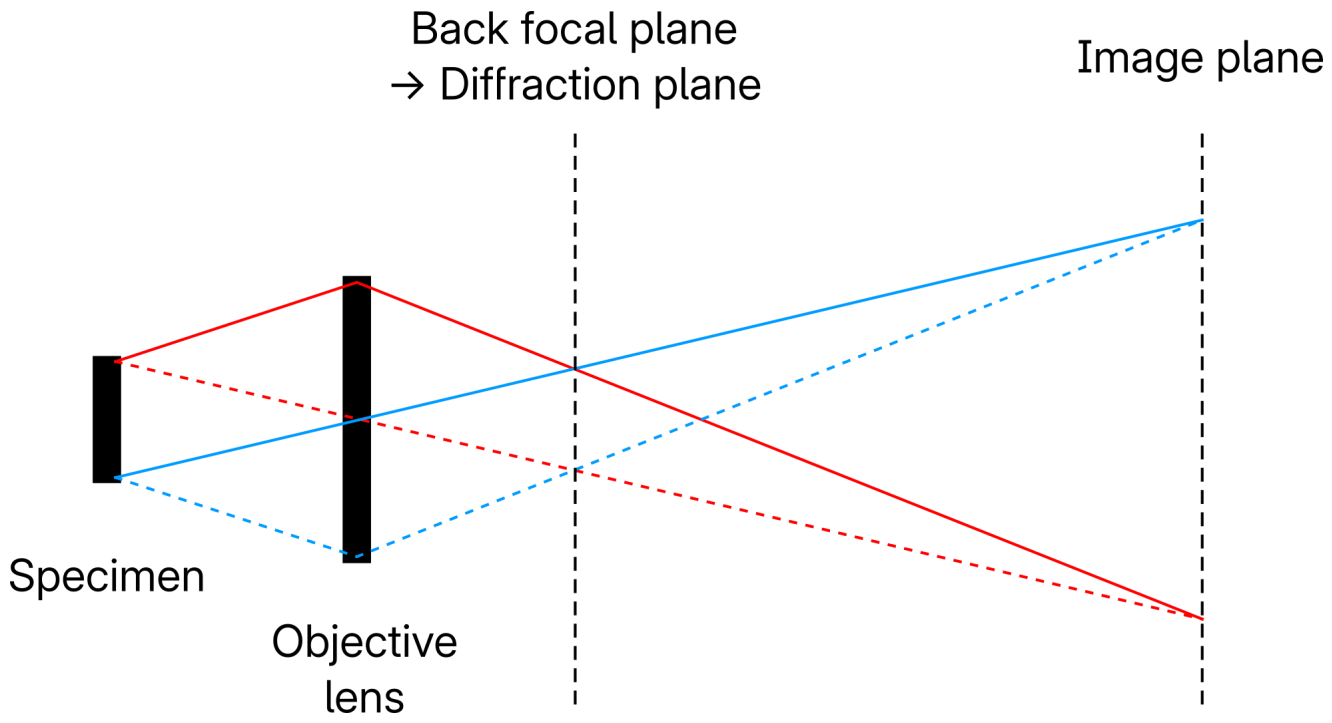


(S)TEM Geometry: Image plane



The wave at the image plane is (ideally) identical to the exit wave of the specimen

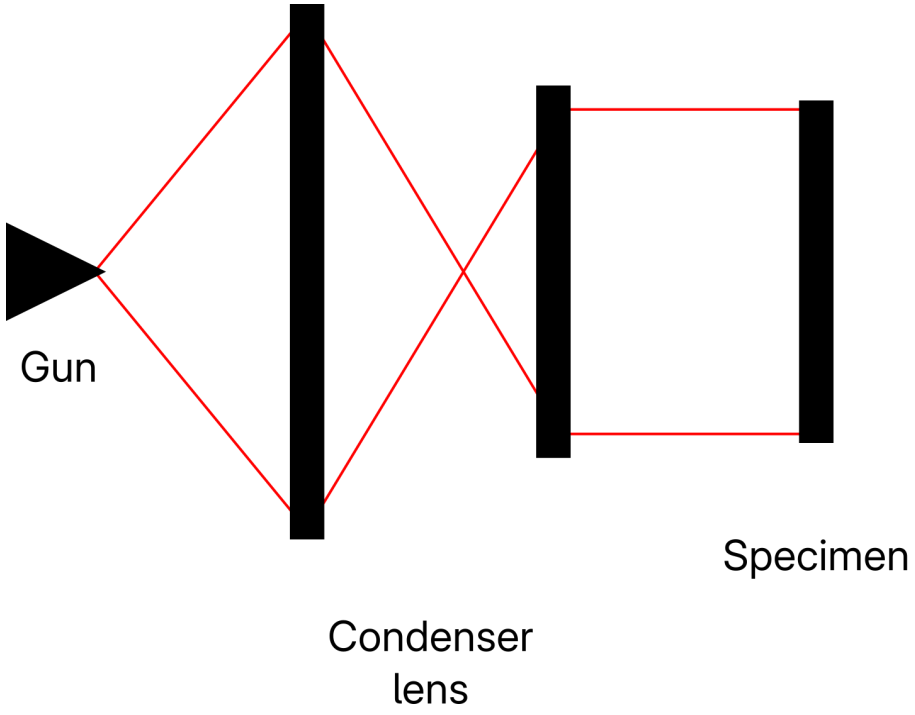
(S)TEM Geometry: Diffraction plane

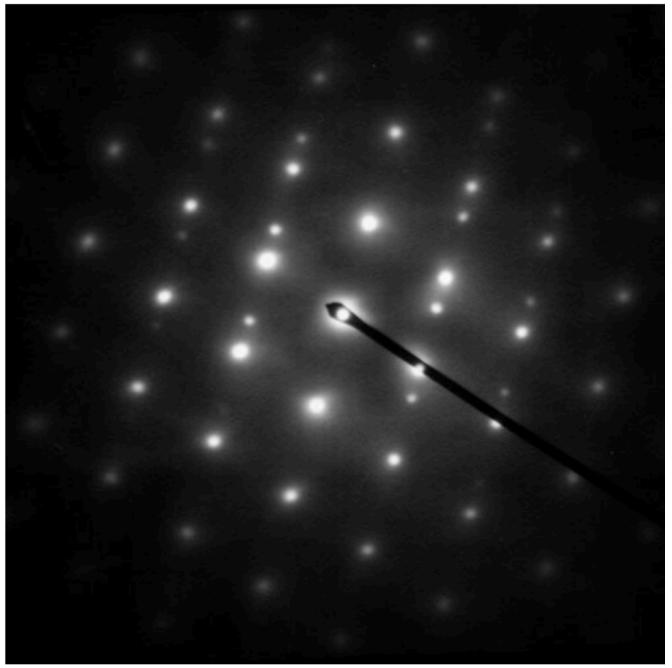


Back focal plane = Image plane of infinity

→ Fraunhofer (far-field) diffraction = Fourier transform = Reciprocal space!

(S)TEM Geometry: Parallel illumination

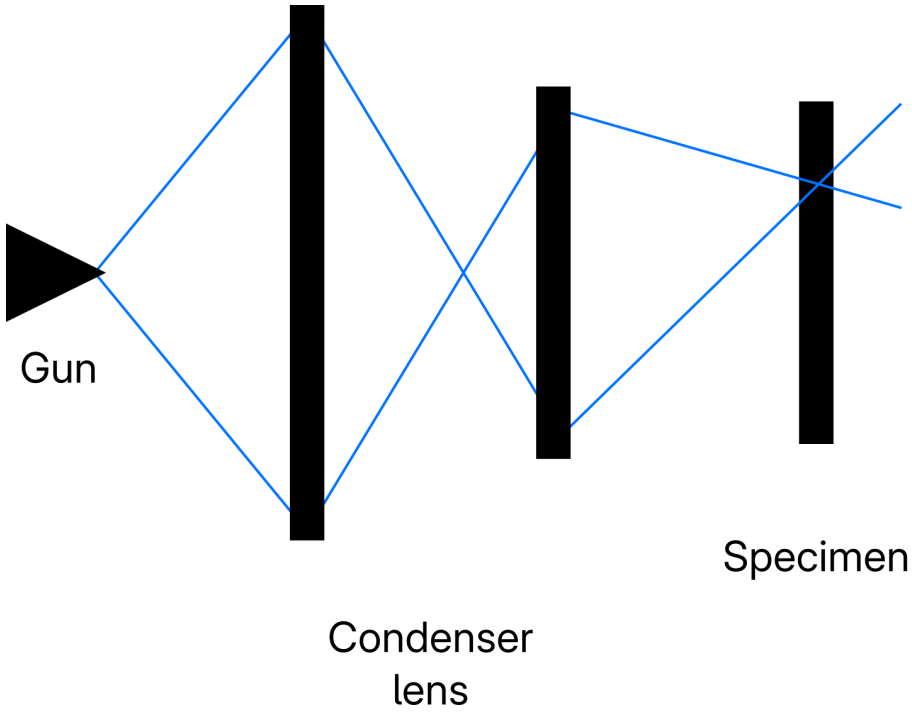


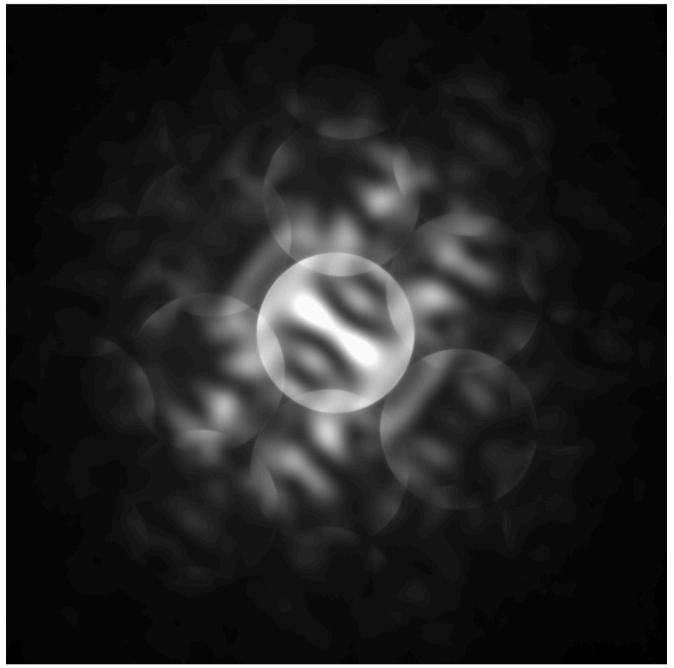


Parallel beams create spot patterns → Selected Area Diffraction Pattern (SADP)

Zone Axis Diffraciton Pattern of Twinned Austenite in Steel. From https://commons.wikimedia.org/wiki/File:Austenite_ZADP.jpg.

(S)TEM Geometry: Convergent beam illumination





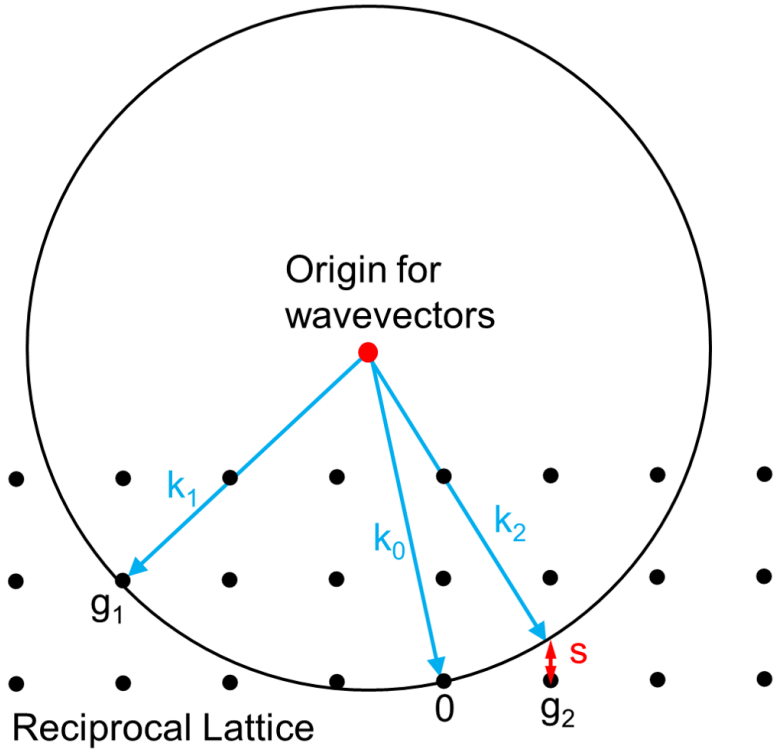
Aperture decides convergence angle (α)

Convergent beams create “Bragg discs” → Convergent Beam Electron Diffraction (CBED)

CBED pattern of GaN/InGaN. From https://www.gatan.com/resources/media-library/110-cbed-pattern-ganingan-0?modal=1

Ewald sphere

Diffraction pattern → Intersection of Ewald sphere and reciprocal lattice



https://commons.wikimedia.org/wiki/File:Ewald3.png

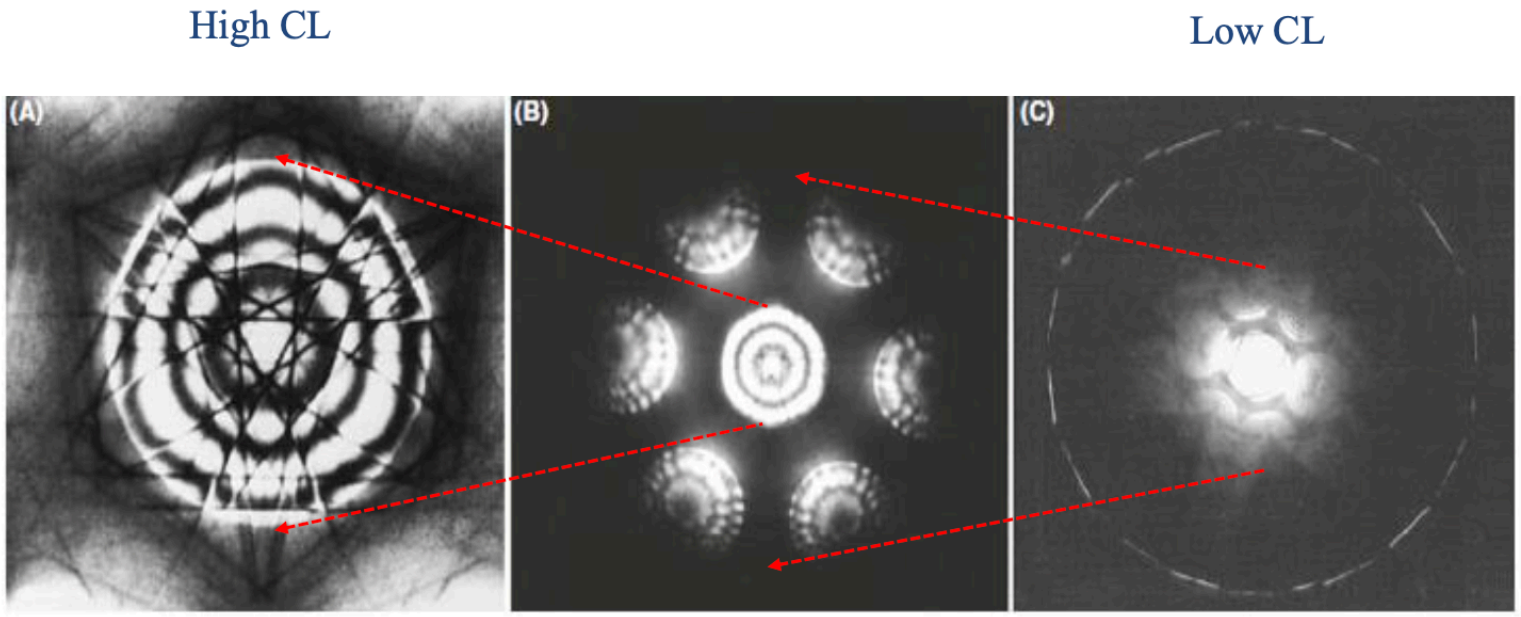
Ewald sphere

Thin sample in z direction → Projection of reciprocal lattice, “relrods”

$$\psi(x,y)\cdot\delta(z) \quad \leftrightarrow \quad \Psi(k_x,k_y)\cdot 1$$

Laue Zones

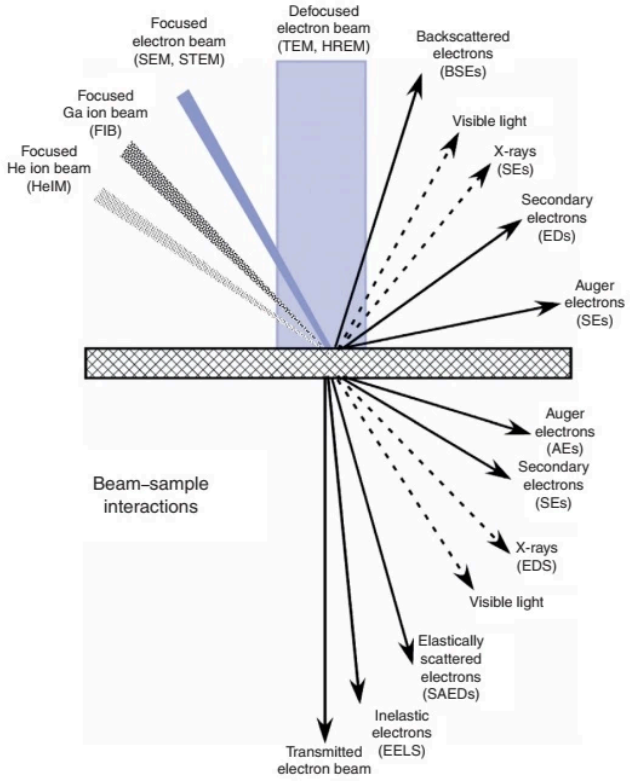
Laue zones (LZ): diffraction planes perpendicular to the incident beam



Convergent beam rotates the Ewald sphere → We can find Higher-Order LZ (HOLZ) in CBED!

What is an exit wave?

Some function of incident beam (illumination) ψ_{in} and the sample potential V



https://vaccoat.com/blog/transmission-electron-microscopy-tem/

Projection approximation

$$\psi_{\text{exit}}(\mathbf{x}) = \psi_{\text{in}}(\mathbf{x}) \cdot e^{i\sigma V(\mathbf{x})}$$

cf)

$\psi \sim e^{iS/\hbar}$

$\Delta\phi = (1/\hbar)\int (eV - e\mathbf{v}\cdot\mathbf{A})dt$

Thicker specimens have dynaimcal scattering

Strongly interacting or thicker samples scatter electrons more than once.

- Diffracted beams couple to one another
- Intensity depends strongly on thickness and orientation
- A focused probe can channel along atomic columns

Modelling interactions:

- Bloch-wave eigenstate method
- Multislice approximation

Why do we need lenses? → Phase problem

In a perfect world, we completely know the linear propagation operator $\mathcal{P}$

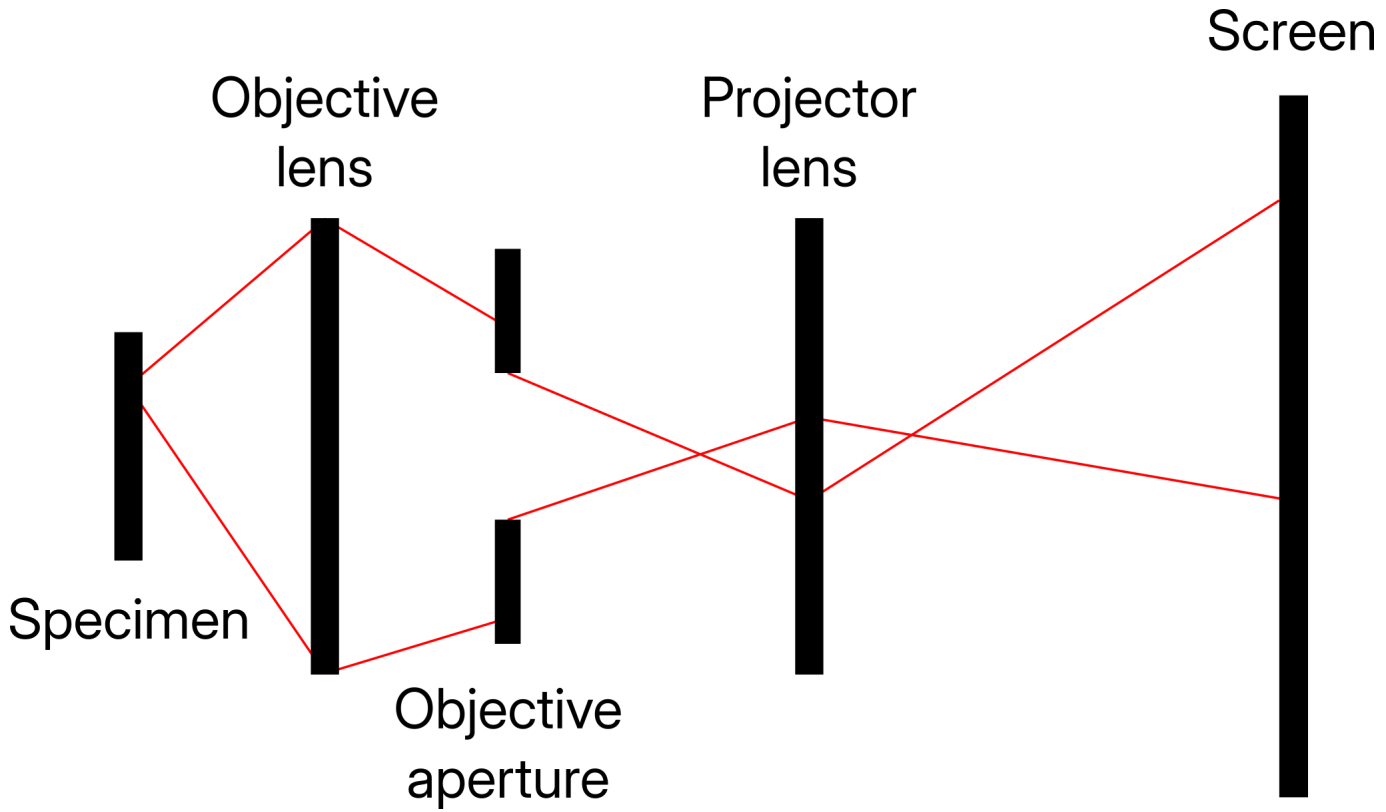
$$\psi_{\text{screen}} = \mathcal{P}[\psi_{\text{exit}}]$$

Still, we can only measure the intensity of the electron wave:

$$I = |\psi_{\text{screen}}|^2$$

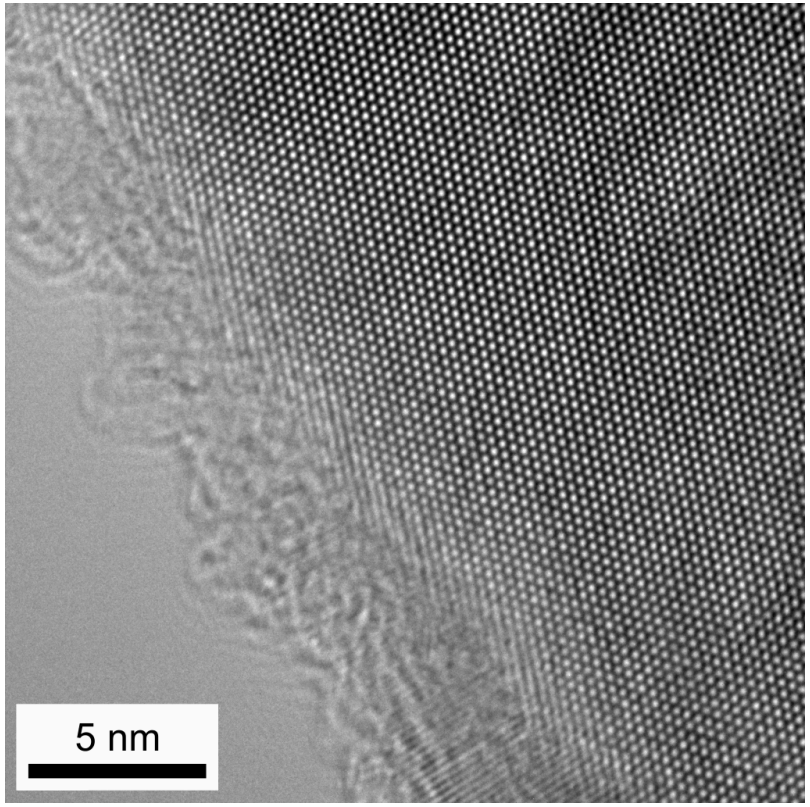
We need the objective lens(es) to focus back to an real-space image

Exit wave → Detector



TEM imaging

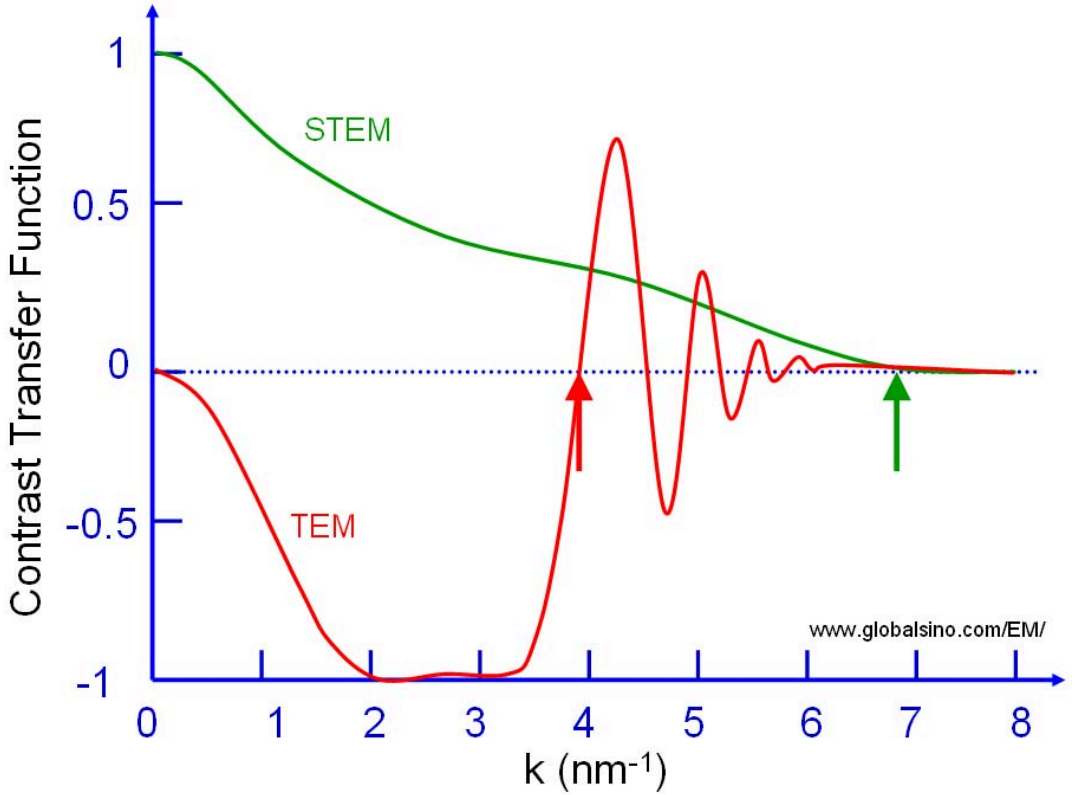
The conventional imaging method



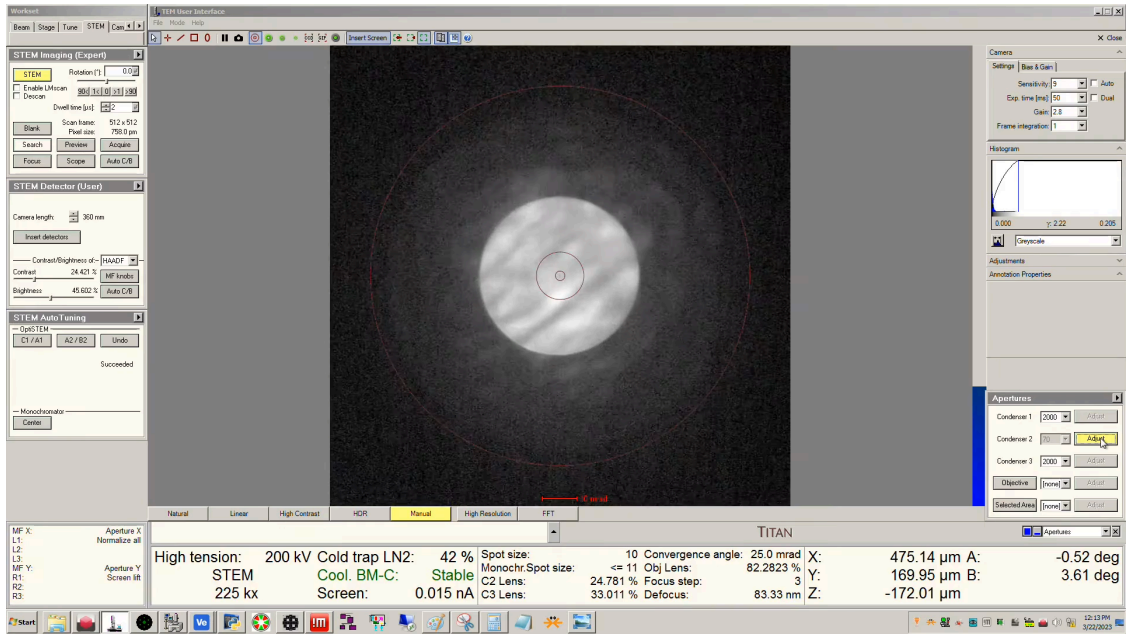
HRTEM image of magnesium. From https://commons.wikimedia.org/wiki/File:HrtemMg.png

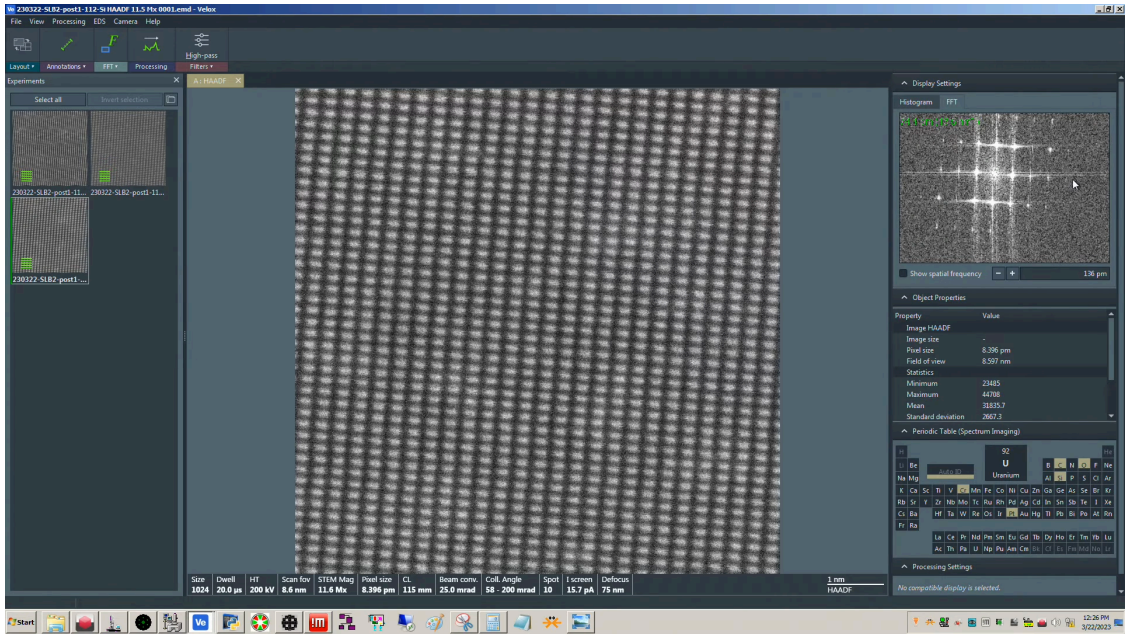
Contrast Transfer Function, Point Transfer Function

Lenses also have their own transfer functions:



STEM imaging



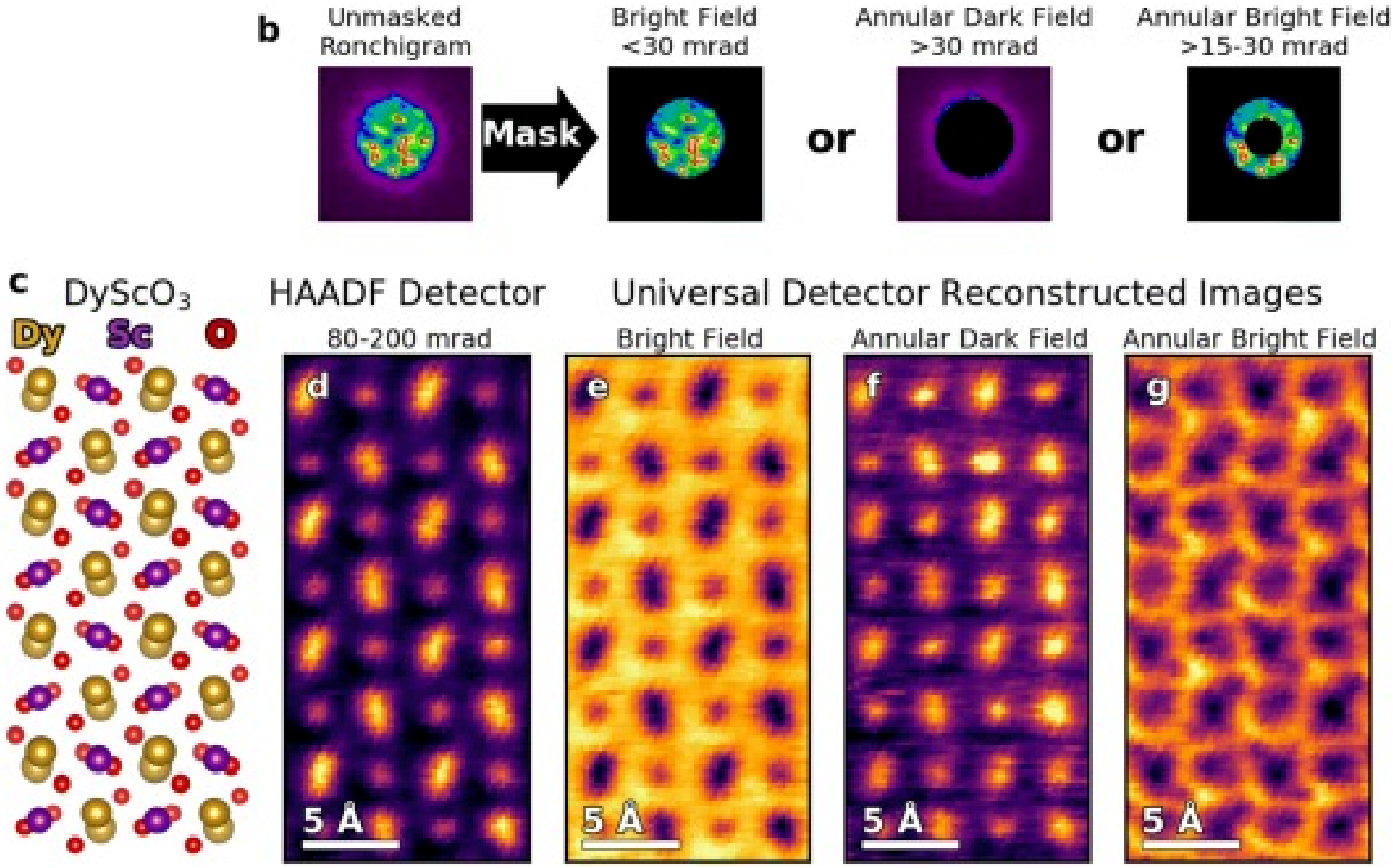


1 pixel per probe position → 2 displays: 1 for beam and 1 for scanned “image”

Screen does not contain image! Instead it contains rich information of specific probe position

Conventional STEM : Brightness of pixel = # of electrons hitting detector

Detecting parts of the CBED



Hachtel et al. (2018). https://doi.org/10.1186/s40679-018-0059-4

Pixelated detectors: 4D-STEM

Specimen plane (~ real space) probe positions $\{R_x, R_y\}$

$\otimes$

Detector plane (~ reciprocal space) pixel positions $\{k_x, k_y\}$

→ 4-dimensional array

Roadmap

- TEM
  - 4D-STEM
    - Ophus, Four-Dimensional Scanning Transmission Electron Microscopy (4D-STEM): From Scanning Nanodiffraction to Ptychography and Beyond (2018) https://doi.org/10.1017/S1431927619000497
  - Ptychography
- Materials / PLD
  - Sheplin et al., A practical guide to pulsed laser deposition (2023) https://doi.org/10.1039/d2cs00938b
  - Eason, Pulsed Laser Deposition of Thin Films (2006)